

Molecular design software is now free

Building a browser molecular workbench with GPT-5.6 Sol

Rafal P. Wiewiora Woody Sherman

PsiThera, Watertown, MA

molarium.org August 2026

Abstract

What happens when a chemist asks a coding agent for the molecular-design environment they actually want, then refuses to accept toy chemistry? Over several days, that conversation produced MOLARIUM: a public browser workbench for three-dimensional editing, protein preparation, conformer search, molecular mechanics, ANI-2x, and short-chain protein prediction. GPT-5.6 Sol did not merely connect existing commands. It wrote browser-native Sage forces, OBC2/ACE solvent, SHAKE/RATTLE constraints, a homogeneous-replica molecular dynamics engine, and ANI-2x descriptor and force kernels. We tested those parts against OpenMM, TorchANI, finite differences, and fixed fixtures. The artifact is therefore more than another software release. It is evidence that, under expert direction, a current model can turn published methods into substantial, inspectable scientific code while the research conversation is still under way. Equations and parameters made the new implementations testable; trained weights and accumulated validation still had to be preserved and reused.

The result is not merely that an agent can operate scientific software. A chemist asked for missing software, tested it while it was being written, and kept the resulting methods as inspectable code.

1 A molecule should not disappear into its software

Molecular design begins with an act of spatial imagination: a chemist sees a molecule that does not yet exist. It is a beautiful job, and its software should honor it. Instead, the idea is often broken into a drawing package, a preparation wizard, a simulation script, a queue, and a different viewer for the answer. Each handoff asks the chemist to remember what the software has forgotten. The tools become the work.

We wanted the opposite feeling: the quiet pleasure of software that simply works. The molecule should turn when the chemist turns it. A carbonyl-to-enol edit should remain in front of them until both bond-order changes are finished. A calculation should return to the same scene, not to an anonymous output directory. The machinery should recede so that attention can stay with shape, strain, water, and the next chemical idea.

Why the browser, and why now. For many years the browser was mainly a viewer or a front end to a remote calculation. Two platform changes made a local scientific application plausible. WebAssembly provided a portable, sandboxed target for mature C and C++ libraries, avoiding a complete rewrite in JavaScript [1]. WebGPU then exposed modern GPU buffers, compute shaders, and command queues through one browser API backed by native graphics systems [2]. Web molecular graphics had already established the interactive side of the problem [3]. The remaining step was to place chemical perception, numerical kernels, and the molecular canvas in the same local runtime.

This division suits molecular design. WebAssembly handles graph operations, file formats, control flow, and reference calculations. WebGPU handles repeated floating-point work across atoms or conformers. Workers keep long calculations off the interface thread. The application can be served as static files, while the structures and supported calculations remain on the user’s device. OpenMM also provides an independent reference against which narrower browser kernels can be tested [4].

The more provocative result is not the interface but how it appeared. The first instruction was almost unserious: build a copy of an existing molecular viewer. Minutes later the first benzene was visibly wrong. “Can you actually test all this out?” became the governing question. A methyl fragment made a false bond; a trifluoromethyl group arrived flat; changing one bond order prematurely bent an oxygen and rotated a ligand. Each visual objection became a test and then a change in the model of the molecule. Bond perception was separated from drawing; fragments received three-dimensional starting conformations; coupled edits were staged; cleanup was restricted to the atoms near the change.

The same pattern drove the numerics. “Is this a real force field?” killed an early toy evaluator and moved the project to RDKit MMFF/UFF and prepared OpenFF Sage Systems [5, 8, 9, 10]. A disagreement between CPU and GPU energies exposed both a unit mistake and an architectural misunderstanding: OpenMM Reference in

WebAssembly was a CPU oracle, not the WebGPU engine. “Wait—aren’t we using WebGPU?” led to analytic WGSF forces, then compact sparse buffers, then tests on Trp-cage and ubiquitin. Asking for implicit solvent led to OBC2/ACE [11, 12]; asking whether that included SHAKE led to fixed-point SHAKE/RATTLE [13, 14]; a 100,000-step run that stopped immediately exposed a forgotten 5,000-step validation cap. This was not a model producing a finished application from one prompt. It was a chemist noticing the wrong thing on screen and an agent turning that objection into code, repeatedly.

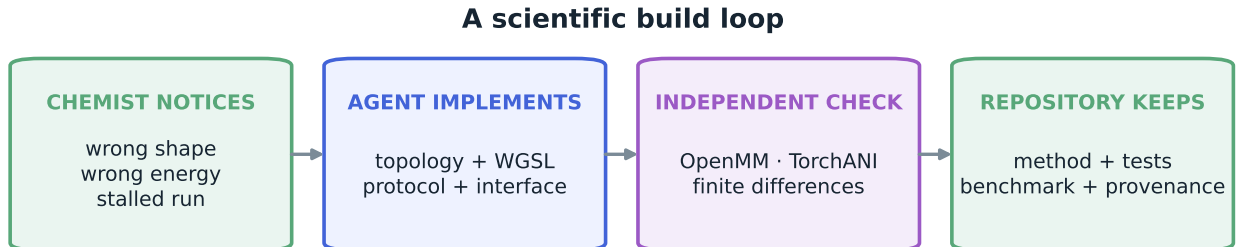


Figure 1: The development conversation became a repeatable scientific loop. Visual and numerical failures produced code changes; independent oracles decided whether those changes survived; accepted methods remained with their tests and provenance.

That history gives “free” four distinct meanings:

- | | |
|---|--|
| <p>1. Free to use. The public artifact has no license fee, and supported calculations need not upload a structure.</p> <p>3. Free to preserve. A repository can keep agent-written methods long enough for other scientists to inspect, test, and improve them.</p> | <p>2. Free to rebuild. <i>We now take seriously the possibility that a sufficiently specified method can be implemented from its paper during a project, rather than awaited as a product.</i></p> <p>4. Free to compose. One interface and explicit protocols can make methods plug-and-play for chemists and for agents running reproducible benchmarks.</p> |
|---|--|

This is different from asking a model to call a fixed toolchain. Orchestration is right when trusted programs and trained models already exist, as recent protein-design systems demonstrate [23, 24]. Accordingly, MOLARIUM keeps RDKit, OpenMM, ONNX Runtime, and the official ANI-2x and OpenFold weights [5, 4, 18, 16, 17, 20]. GPT-5.6 Sol wrote the missing browser-specific layers. Classical mechanics is unusually amenable to this division: equations, parameters, units, and reference values can make a new implementation answerable to independent calculation. A neural architecture alone cannot recreate its weights or its biological evidence. The model identity, conversation, commits, and validation outputs are part of the project record [22].

2 The experiment as it unfolded

By the end of the conversation, opening MOLARIUM felt less like launching a suite than picking up one persistent molecular model. The 7KPA complex [21] in Figure 2 can be prepared, inspected, edited, and locally relaxed without leaving its scene. The preparation path resolves alternate locations and explicit connectivity, repairs supported residue atoms, adds hydrogens, and keeps a crystallographic water when its geometry supports a ligand-protein bridge. It does not conceal unresolved chemistry behind a green button.

The continuity is practical, not decorative:

- **View** keeps chains, ligand, waters, contacts, and the selected residue together.
- **Build** stages bond-order, charge, and hydrogen edits before cleanup is allowed to move the structure.
- **Inspect** returns aligned trajectories to the same canvas while preserving the raw coordinates for export.



Figure 2: The same 7KPA protein–ligand system during (A) contact inspection, (B) staged chemistry and pocket-local editing, and (C) residue-focused inspection. The molecule, selection, and chemical context survive the change of task.

3 What the browser had to learn

The browser is liberating precisely because it refuses the usual assumptions. There may be no Python, compiler, CUDA driver, or filesystem path that survives the next machine. WebAssembly gives mature C/C++ code a portable sandbox; WebGPU gives the page buffers, compute shaders, and command queues, but only fast f32 arithmetic and no portable 64-bit atomics. The solution was not to pretend that these constraints did not exist. It was to make them the architecture.

Chemical preparation ends at a numeric System: explicit masses, charges, bonded terms, Lennard–Jones parameters, exceptions, constraints, and optional generalized Born parameters. The GPU kernels never guess atom types. Browser small molecules use OpenFF Sage 2.1; the bundled protein fixtures preserve exact numeric Systems exported from OpenFF Rosemary 3.0.0-alpha0 with NAGL charges [10, 19]. From that common boundary, the code chooses a layout that matches the scientific workload:

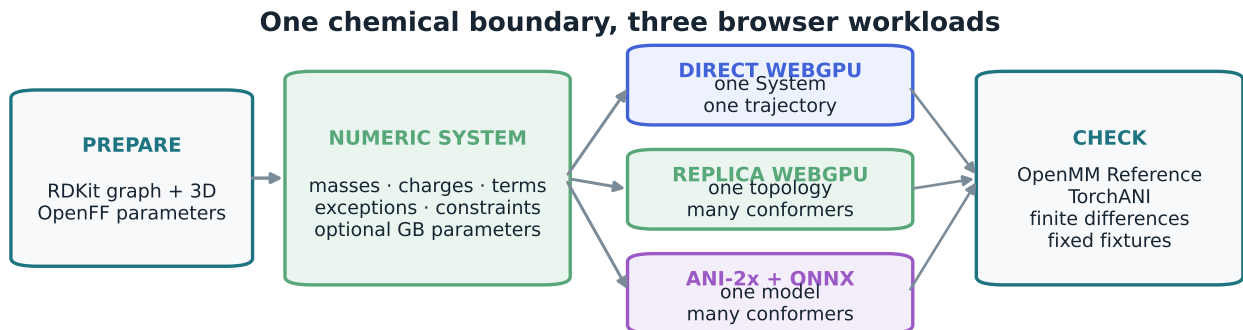


Figure 3: Preparation stops at an explicit numeric System. Three local execution layouts consume it or its coordinates, and each has a deliberately separate reference path.

- **One System, one trajectory.** Atom-centric WGS� kernels evaluate analytic Sage valence, Coulomb, Lennard–Jones, and OBC2/ACE forces from sparse incidence buffers [10, 11, 12].
- **One topology, many replicas.** A second engine adds a replica dimension to every kernel, borrowing STORMM’s synthesis idea rather than its source code [15]. Split-int64 force accumulation and fixed-point coordinates replace missing 64-bit atomics; replica-local SHAKE/RATTLE permits a 2 fs Langevin step [13, 14].
- **One model, many conformers.** WGS� constructs ANI-2x atomic environment vectors, ONNX Runtime evaluates the official eight TorchANI networks, and WGS� contracts derivatives back to coordinate forces without a CPU round trip [16, 17, 18].

Each path has a deliberately narrow contract. The replica engine accepts homogeneous topologies, nonperiodic implicit solvent, and no PME; long-time stability remains to be established. The ANI-2x path is a vacuum model for neutral, closed-shell molecules of at most 96 supported atoms. OpenFold 2 is the opposite case: its scientific content is in trained weights, so the browser runs the existing model locally rather than claiming to reconstruct it [20].

Equations can be reconstructed and checked. Trained weights, parameter sets, and accumulated experimental evidence must be preserved.

Every fast path answers to slower code. OpenMM 8.2 Reference WebAssembly re-evaluates the same numeric System in double precision [4]; native TorchANI checks ANI; finite differences check forces. Reference code is a test instrument, not a user-facing fallback. This distinction mattered throughout: agreement checks arithmetic, but shared parameters do not independently validate force-field assignment.

4 The conformer experiment

Conformer search became the first place where all of these choices met. The useful result was not one photogenic minimum but a diverse, inspectable set whose provenance stayed visible.

ETKDGv3 seeds; MMFF94/UFF polish; Sage replica dynamics; common Sage and ANI scores; symmetry-aware clustering.

RDKit generates up to 64 symmetry-pruned ETKDGv3 seeds [5, 6, 7] and briefly polishes them with MMFF94 or UFF [8, 9]. The replica schedule then applies 2,500 Langevin steps at 600, 450, and 300 K, followed by 240 fixed-step Sage relaxation iterations in OBC2/ACE [10, 11, 12]. This final stage is a bounded relaxation, not a converged minimizer. Symmetry-aware heavy-atom RMSD removes duplicate structures before clustering and SDF export.

The comparison retains three generators from the same seeds: MMFF-polished coordinates, replica/Sage coordinates, and ANI-2x-refined coordinates. All final coordinates receive a common Sage/OBC2 energy and, for supported molecules, a vacuum ANI-2x energy. The common Sage score makes the generators comparable and checks replica-engine parity on identical coordinates. It is not an independent physical oracle: the replica lane optimizes the same objective. The map can plot energy, structural variables, or rank against rank; color reports the generator, not the scoring model.

In an eight-molecule development panel, replica search found a lower common Sage minimum for four molecules. The median best-energy gain was only 0.128 kcal/mol. It added 75 structural clusters, including 44 within 3 kcal/mol that were absent from the seed lane’s window. The current evidence supports faster exploration and added diversity. It does not establish better conformer quality in general.

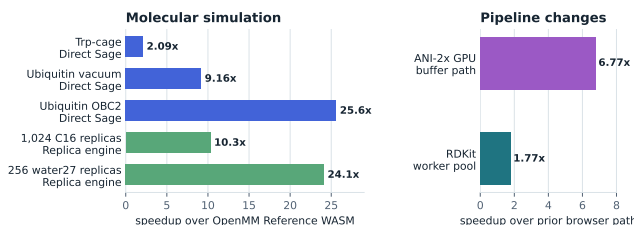


Figure 4: Engineering timings on one Apple M1 Pro. Direct and replica simulations use OpenMM Reference WebAssembly baselines; replica results omit setup and readback and use different integrators. ANI-2x and RDKit compare with prior browser paths. None are native-CPU or CUDA comparisons.

5 What survived measurement

The first honest result was negative: a GPU is a poor place to launch one tiny molecule. OpenMM Reference was 13.3-fold faster for one C16 and 3.5-fold faster for one water27 because GPU dispatch dominates tiny jobs. Larger or batched work reversed the result. Direct WebGPU reached 2.09-fold the Reference throughput for 304-atom Trp-cage, 9.16-fold for 1,231-atom vacuum ubiquitin, and 25.6-fold for ubiquitin with OBC2/ACE. At equal aggregate steps, 1,024 C16 replicas reached 10.3-fold and 256 water27 replicas 24.1-fold the Reference throughput. The baseline is double-precision OpenMM Reference WebAssembly, not native CPU or CUDA. Replica timings exclude setup and readback, and the integrators differ.

WebGPU did not make one small molecule fast. It made enough molecular work fast.

Numerical tests are separate from product regressions. Direct Sage’s largest relative energy deviation from OpenMM was 2.361e-6; aspirin force relative RMS was 7.696e-7. Aspirin OBC2 differed by 1.279e-4 kcal/mol with 7.835e-7 force relative RMS. The replica engine’s largest energy error was 2.15e-5 kcal/mol and force relative RMS

was 6.66e-6. Against native TorchANI on six small molecules, ANI-2x reached 0.02572 kcal/mol maximum energy error and 1.022e-5 force relative RMS. These fixtures test implementation parity, not force-field accuracy, ANI transferability, or cross-vendor agreement.

The 418 browser checks prevent product regressions; they are not 418 scientific validations. Against prior browser paths, GPU-resident ANI descriptors and forces improved that stage 6.77-fold, while parallel RDKit workers improved a cold seed-generation run 1.77-fold (Figure 4).

6 What this does—and does not—show

The engines are nonperiodic and omit PME, barostats, membranes, and arbitrary OpenMM plugins. The single-precision, all-pairs replica engine accepts one topology at a time. OBC2 and browser protein parameterization remain experimental; ANI-2x is a restricted vacuum model; OpenFold supports one short chain and uses a network service only for a user-requested MSA.

Next tests require a frozen conformer benchmark with repeated seeds, reference-conformer recall, an independent energy judge, confidence intervals, and a matched modern CPU baseline. The engines also need a cross-GPU matrix and longer NVE and constrained-solvent trajectories.

The conversation is part of the evidence, not decoration. We are preserving its timestamped transcript alongside commits, test outputs, benchmark artifacts, and file hashes, with credentials and private structure data removed before public release. Until active time is reconstructed from that archive, “several days” denotes the observed project interval rather than a controlled productivity measurement.

MOLARIUM is public at <https://molarium.org>; source is at <https://github.com/rafwiewiora/molarium>. We began by asking for a nicer place to look at molecules. We ended with a sharper proposition: a chemist can now ask for missing scientific software during the work itself, watch it fail, demand the relevant comparison, and keep the resulting method as a public, testable artifact. GPT-5.6 Sol supplied extraordinary implementation speed. The chemist supplied the taste to notice a flat trifluoromethyl group, the suspicion to question an energy scale, and the judgment to decide which failure mattered. Neither role is dispensable. The exciting future is not chemistry without chemists; it is chemistry in which beautiful ideas no longer have to wait for beautiful tools.

References

- [1] A. Haas et al. WebAssembly. *PLDI* (2017).
- [2] W3C GPU for the Web Working Group. [WebGPU specification](#).
- [3] A. S. Rose and P. W. Hildebrand. NGL Viewer. *Nucleic Acids Research* 43, W576–W579 (2015).
- [4] P. Eastman et al. OpenMM 8. *J. Phys. Chem. B* 128, 109–116 (2024). doi:10.1021/acs.jpcb.3c06662.
- [5] G. Landrum. RDKit: open-source cheminformatics, Release 2025.03.4. doi:10.5281/zenodo.591637.
- [6] S. Riniker and G. A. Landrum. Better informed distance geometry for conformer generation. *J. Chem. Inf. Model.* 55, 2562–2574 (2015). doi:10.1021/acs.jcim.5b00654.
- [7] S. Wang et al. ETKDG conformer generation for small rings and macrocycles. *J. Chem. Inf. Model.* 60, 2044–2058 (2020). doi:10.1021/acs.jcim.0c00025.
- [8] T. A. Halgren. Merck molecular force field I: MMFF94. *J. Comput. Chem.* 17, 490–519 (1996).
- [9] A. K. Rappé et al. UFF. *J. Am. Chem. Soc.* 114, 10024–10035 (1992). doi:10.1021/ja00051a040.
- [10] S. Boothroyd et al. Open Force Field 2.0.0: Sage. *J. Chem. Theory Comput.* 19, 3251–3275 (2023). doi:10.1021/acs.jctc.3c00039.
- [11] A. Onufriev, D. Bashford, and D. A. Case. Modified generalized Born model. *Proteins* 55, 383–394 (2004). doi:10.1002/prot.20033.
- [12] A. Schaefer and M. Karplus. Analytical continuum electrostatics. *J. Phys. Chem.* 100, 1578–1599 (1996). doi:10.1021/jp9521621.
- [13] J.-P. Ryckaert, G. Ciccotti, and H. J. C. Berendsen. SHAKE constraints. *J. Comput. Phys.* 23, 327–341 (1977). doi:10.1016/0021-9991(77)90098-5.
- [14] H. C. Andersen. RATTLE. *J. Comput. Phys.* 52, 24–34 (1983). doi:10.1016/0021-9991(83)90014-1.
- [15] D. S. Cerutti, R. Wiewiora, S. Boothroyd, and W. Sherman. STORMM. *J. Chem. Phys.* 161, 032501 (2024). doi:10.1063/5.0211032.
- [16] X. Gao et al. TorchANI. *J. Chem. Inf. Model.* 60, 3408–3415 (2020). doi:10.1021/acs.jcim.0c00451.
- [17] C. Devereux et al. ANI-2x: extending ANI to sulfur and halogens. *J. Chem. Theory Comput.* 16, 4192–4202 (2020). doi:10.1021/acs.jctc.0c00121.
- [18] Microsoft. [ONNX Runtime](#), Web release 1.27.0.
- [19] Open Force Field Initiative. OpenFF Rosemary 3.0.0-alpha0 and NAGL openff-gnn-am1bcc-1.0.0; exact fixture provenance is archived with [Molarium](#).
- [20] G. Ahdriz et al. OpenFold. *Nature Methods* 21, 1514–1524 (2024). doi:10.1038/s41592-024-02272-z.
- [21] Protein Data Bank. Structure 7KPA. doi:10.2210/pdb7KPA/pdb.
- [22] OpenAI. GPT-5.6 Sol. Model-assisted development record, prompts, commits, test artifacts, and hashes are described in the [Molarium provenance archive](#) (2026).
- [23] Anthropic. [Claude accelerates protein design](#) (2026).
- [24] OpenAI. [Introducing new capabilities to GPT–Rosalind](#) (2026).